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TCE–602

B. TECH. (CE) (SIXTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

RCC—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) All question are compulsory.

(ii) Answer any *one* sub-question among (a) and (b) in each main question.

(iii) Each sub-question carries 10 marks.

1. (a) Explain the difference between pre-tensioning and post-tensioning. (CO1)

OR

(b) Explain the different step of design for flexure. (CO1)

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2. (a) Describe the various methods of pre-stressing. (CO1)

OR

(b) Determine the various stresses set up at mid span in a pre-tensioned beam 250 mm × 500 mm, subjected to an initial pre-stress of 1500 kN and a uniformly distributed superimposed load of 5 kN/m over a span of 15m. Assume total loss of pre-stress as 12% and eccentricity of pre-stress at mid span 100 mm. (CO1)

3. (a) Calculate the percentage loss of pre-stress due to shrinkage of concrete in a beam pre-stressed by a calbe carrying an initial pre-stress of 600 N/mm^2. The pre-stress is transferred at the age of 7 days. Assume the beam as : (CO1, CO2)

(i) Pre-tensioned

(ii) Post-tensioned

OR

(b) Draw the bending moment envelope after 18 % redistribution, for a fixed ended beam of span 6 m carrying 25 kN/m load at collapse. (CO1, CO2)

4. (a) What do you understand by term "redistribution of moment" ? Explain the conditions in which IS code method is applicable. (CO2)

OR

(b) Draw the bending moment envelope after 30% redistribution, for a fixed ended beam of span 5m carrying 20 kN/m load at collapse. (CO2)

5. (a) Design a rectangular beam, continuous over three column supports of effective span 7 m. The beam is subjected to an imposed load 13 kN/m and live load of 15 kN/m. Use M20 concrete and Fe415 steel. (CO2)

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OR

(b) Design a continuous rectangular beam of span 6 m to carry a dead load of 12 kN/m and live load of 16 kN/m. The beam is continuous over the three span and simple supported by a column. Use M20 concrete and Fe 415. Draw the bending moment envelope after redistribution. (CO2)

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